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Why Less Praise for Enhanced Performance?

*Moving Beyond Responsibility-Shifting, Authenticity, and
Cheating Toward a Nature-of-Activities Approach*

FILIPPO SANTONI DE SIO, NADIRA S. FABER,

JULIAN SAVULESCU, AND NICOLE A VINCENT

Introduction

Good performance often attracts praise. But suppose that Llana improves her performance on a Latin language test by using modafinil or transcranial direct current stimulation (tDCS) to enhance her ability to memorize new words (e.g., see Gilleena et al.¹; or Meinzer et al.²). Would Llana be due less praise than if she had obtained the same grade but without cognitive enhancement?

Those who endorse the *less praise intuition* (LPI)—for instance, lay people in our own and in others' studies—might appeal to responsibility-shifting, authenticity, or cheating arguments to support the intuition that less praise is indeed due to people like Llana. However, we draw on examples from performance enhancement in sport and professional contexts to demonstrate that these arguments leave something out, and then we develop a better justification for LPI. On our account, praise is diminished not because it is shifted to someone else, because it is due to an inauthentic self, or because an otherwise good performance is blemished by cheating, but because enhanced actors engage in very different activities; because of this, we need different yardsticks to assess their performance.

This chapter offers a novel perspective on the issue of praise in the presence of cognitive enhancement,ⁱ but it also presents an outline of a methodology for ethical reflection on performance enhancement more generally.ⁱⁱ

LPI and the Negative View of Enhancement: Empirical Data

Based on previous research (e.g., Caviola et al.⁶, Faber et al.⁷, Faulmüller et al.⁸, Schelle et al.⁹), we thought it likely that LPI may be fueled by the public's negative attitude toward pharmacological performance enhancement. In contrast to previous studies that presented hypothetical scenarios to the participants (e.g., Scheske and Schnall¹⁰ or subtly alerted participants to possible concerns (e.g., Forlini and Racine¹¹), we wanted to investigate which concerns over the use of enhancement lay people raise (1) given the actual current status of enhancement in society (i.e., its current medical development, social distribution and acceptance, existing legislation, etc.) and (2) without being prompted to any specific concern in advance.

To this end, we conducted a survey in a controlled laboratory setting that included 102 university science students (72 female, 30 male) with a mean age of 22.7 years. Our questionnaire contained a neutrally phrased description of pharmacological enhancement, which stated that some medical substances that were initially developed to treat disease can also improve performance in healthy individuals. To avoid raising concerns that may otherwise not have occurred to our participants, we stated our questions in as indirect and neutral a manner as possible: "In what kind of situations should healthy people take performance-improving substances, if any? If they should not take them, please explain why."

Participants were handed the questionnaire and were asked to write down their responses. To ensure coding reliability, participants' written answers were coded by two trained independent coders and then analyzed using a content analysis technique that arranged answers into clusters of similar content; the frequency of answers in each cluster was then determined. The two coders concurred in 90% of cases; in the remaining 10% of cases, they reached agreement after discussion. Using this qualitative research approach, we cannot ascertain the causal relationship between specific concerns—for instance, whether competitive fairness would still be an issue even if distributive fairness could be assured. However, we can assess whether, given the current social status of enhancement, lay people care about a certain concern in the first place (e.g., if they mentioned fairness at all) and thus get a sense of the relative prevalence of various concerns.

Our overall finding confirms that *lay people have a generally negative view about enhancement*: even though they were asked about situations in which the use of enhancement might be accepted, 34% of participants responded that no such situation exists. The remaining 66% described at least one situation where enhancement might be acceptable, but the vast majority of these

situations were characterized by absences of potential concerns. For example, several participants responded that enhancement might be acceptable if the situation were *not competitive* or if the substance had *no physical side effects*. Underlying concerns could thus be identified from these seemingly positive answers.

Even without prompts, participants raised a range of concerns about enhancement. The most commonly cited concern related to the unintended side effects of enhancers on the user's health or on their behavior (60% of participants). Fairness and/or competition and/or cheating were mentioned by 54% of our participants. (Because folk reasoning is not as fine-grained as philosophical arguments, we cannot distinguish between these three related factors in our data.) Other concerns included the risk of addiction or dependence (11%), whether the use of enhancers entailed breaking laws or rules (9%), and whether enhancement might undermine users' confidence in their own abilities (7%). No significant correlations between these clusters were found, suggesting that participants who raised a concern from one cluster were not more likely to also emphasize a concern from another cluster.

Leaving aside concerns about side effects, *cheating* loomed prominent, which is in line with implications of other recent research for different forms of enhancement (Faber et al.¹²). Cheating was mentioned more often than breaking laws or rules, and, given that these two concerns were not correlated, we surmise that worries about cheating cannot be simply restated as worries about rule or law violation. Put another way, enhancement use may still be perceived as cheating even if it is consistent with current rules or laws. And the concern that enhancement might undermine a user's confidence in *his or her own* abilities suggests that lay people have a sense for concerns about ownership of action and/or personal identity, what we here call the responsibility-shifting and authenticity arguments for LPI.

Theoretical Justifications for LPI

Although lay intuitions are sometimes stated in different language, they still nevertheless express the same three concerns as those we mention above—concerns that may be offered as rational justifications for LPI. Here, we demonstrate the form that these concerns take and what role they play in philosophical literature and in public debate about enhancement in sport and education

RESPONSIBILITY-SHIFTING

The responsibility-shifting justification for LPI can be stated as follows: people who enhance their performance through the use of certain substances or

techniques deserve less merit-based praise for their accomplishments because those accomplishments were produced by the enhancers and through the effort of their manufacturers, rather than by the agents' actions and efforts. In the debate about doping in sport, this concern gains expression in the claim that if doping were not disallowed, then we would eventually have competitions between physicians not athletes.

The core intuition here is that enhancers, their producers, or physicians—and, importantly, not athletes—are the real agents in enhanced performance, whereas in nonenhanced performance athletes remain the real agents. But, first, in our view this intuition rests on an implausible metaphysics of action. We find it utterly mysterious why anyone would suppose that enhancement techniques strip agents of their contribution to what they achieve. Second, actual enhancing techniques just do not work like that. Enhanced cyclists must still train hard, they often pedal for hours in races, and this costs them physical and psychological effort. Likewise, enhanced students must still study, expend intellectual effort, write their own assignments, and sit their own tests. Neither enhancers nor enhancer producers do the training, the pedaling, the study, or the writing. Arguments suggesting otherwise rely on a factual mistake about what enhancers actually do (Cf. Mehlman¹³, pp. 492–493).

A milder version of the responsibility-shifting argument is that although enhancers do not literally strip agents of their contribution, they are nevertheless the dominant factors. Here, the claim is not that in sport, for instance, competition between scientists will literally replace competition between athletes, but that technological competition will become a prominent and decisive factor. Consequently, the argument runs, a prominent portion of praise for accomplishment must shift from athletes to scientists. However, in our view, this milder version of the responsibility-shifting argument still rests on the same factual mistake. For instance, steroids merely promote faster recovery from injury and training, but they do not replace any of the human agency. Extraordinary human effort and talent are still the decisive elements in producing outstanding athletic performance. Likewise, cyclists' use of the hormone erythropoietin (EPO) or blood-doping techniques to increase their blood's ability to carry oxygen to the body's cells by increasing the red blood cell count (World Anti-Doping Association¹⁴) does not determine the result of a cyclist's performance any more than altitude training does, regardless of the controversy that these techniques might raise in the media (e.g., Heathers¹⁵ and Mazanov¹⁶).ⁱⁱⁱ They certainly do confer advantage to those who use them over those who do not, but they do not necessarily reduce the contribution of those who use them in terms of their effort, talent, and ability.

AUTHENTICITY

The authenticity justification for LPI is that those who enhance their performance are due no or little merit for their accomplishments because enhancers alter personal identity and thus merit accrues for these accomplishments to different people.

Because, along with others (DeGrazia¹⁷ Juth¹⁸), we find the metaphysical reading of this argument implausible—that enhancers bring about changes in numerical identity, that they temporarily bring new people into existence—we will only discuss a weaker moral reading. On this weaker reading, enhancers that alter people's mind may also create new moral subjects, with their own records of achievements and merits. Morality and law already recognize that substance-induced mental capacity alterations (e.g., alcoholic intoxication) can diminish one's responsibility (e.g., Dimock¹⁹). However, although the analogy with enhancement might initially seem promising, a critical disanalogy is that blame is reduced in alcoholic intoxication cases (when it is; often it is not) when some major mental capacity—typically the minimal capacity for rational thinking and for self-control—falls below some threshold. This is patently not what would happen in cases of enhancement where capacity is increased. Substances affect people's responsibility when they adversely affect their relevant capacities, not because by altering their minds they bring into existence new moral agents with their own records of merit and demerit (e.g., see Vincent²⁰, or Maslen et al.³). Thus, we think that even the weak reading of the authenticity justification fails to justify LPI.

CHEATING

The cheating justification for LPI is that because enhanced agents obtained their achievements either through intentional fraud or through unintentional acquisition of unfair advantage (e.g., as when an athlete unknowingly gains a performance boost from medications legitimately taken to treat a disease), their praise should be reduced either by the amount of blame due for engaging in fraud (e.g., because we think that it blemishes their otherwise good performance) or by the value of the advantage that they gained which others did not (e.g., see Fukuyama²¹)

Yet again, in our view, the cheating concern fails to fully justify LPI. In brief, the fraud version of this concern makes praise reduction dependent on the presence of blame for intentional cheating, but yet intentional cheating is not the only way for athletes to get in trouble. And the unfair advantage version is not completely convincing because it focuses on the ethical implications of the violation of a rule of fairness, which opens it up to the counterobjection

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that absent the ban, absent the cheating. If an unlevel playing field is the only ground for concern, then why not level the playing field by allowing everyone to enhance? (Savulescu et al.²²)

SUMMARY

This section had two explicit aims. One, to convey the gist of the responsibility-shifting, authenticity, and cheating concerns. Two, to cite some reasons why we find these concerns unconvincing.

A New Justification for LPI

Still, in our view, there *is* something to LPI, and this section will elaborate what we think this something might be. We develop an idea that Schermer²³ and Roache²⁴ have voiced about the cheating concern with enhancement—an idea about how activities are defined and characterized—into a novel, broader, and more direct way of justifying LPI. Our approach is novel because, to our knowledge, nobody else has cited, considered, or empirically investigated this idea *qua* justification for LPI. It is broader because it identifies a common thread that ties together and perhaps even explains what really lurks beneath the other three concerns. And it justifies LPI directly rather than via another concern.

THE POINT OF ACTIVITIES

Human activities have different points. Sometimes the point of an activity is an external goal (e.g., earning money through paid work), while at other times its point is internal (e.g., having a meaningful occupation). A combination of internal and external goals is probably most common, but one or the other kind of goal may be prominent in different activities. For instance, share trading is probably predominantly about making money, medicine is predominantly about healing, and the military is predominantly about defense of the realm. This need not prevent stock brokers, surgeons, and military personnel from being fulfilled, vested in, and proud of what they do. Although if they did what they do, but failed to attain those external goals, then criticism would be legitimate. Other activities, however, are predominantly defined through their internal goals. Chatting with a friend is not predominantly about information exchange but about spending time together in a particular way. People run not just to reach a given destination faster than if they had walked but often just to engage in that kind of physical activity. And fiction is read not merely to learn stories but as an activity in itself. Consider someone making the following complaints: “Chatting is such an inefficient, incomplete, and imprecise way

of conveying information!” or “But a motorbike would be way faster!” or “Can’t you just read the synopsis?” An incredulous stare would be a fitting response to someone who so badly missed the point of these activities. We call activities prominently defined through their external goal *goal-directed activities* and activities prominently defined through their internal goal *practice-oriented*.

CHANGING THE NATURE OF AN ACTIVITY

The distinction between *constitutive* and merely *regulative* rules (e.g., Rawls²⁵, Searle²⁶) provides a helpful tool for making our point.^{iv} Namely, some rules are so critical to their activities—because they define and not just regulate them—that changing those rules would involve abandoning the point of those activities. And to the extent (and in the ways) that the new point is less valuable than the old point, this may justify a very distinct kind of complaint as well as a distinct kind of criticism when such rules are infringed (Foddy and Savulescu²⁸).

In goal-directed activities, rules might regulate how external goals should be pursued, but they do not define the nature of those activities. For instance, road rules might stipulate that drivers stay on the right side of the road or that indicators must be used before changing lanes. But someone who drove on the left side or failed to indicate before changing lanes would no less be driving than someone who obeyed the road rules. Rules of such activities can be changed without *the point of those activities being lost*. We think the activities of surgeons, pilots, soldiers, and scientists fall into this category. As long as patients are healed, travelers are brought to their destinations, territories are protected, and scientific breakthroughs are attained, the points of surgery, civil aviation, military, or science (respectively) will still be realized. Such rules may promote safety, efficiency, coordination, or perhaps moral or aesthetic aims (to name just a few examples), but they only regulate not constitute these activities. Consider another example. Musicians’ use of beta-blockers to reduce anxiety and related tremor (Savulescu et al.²²). Since the point here is to put on an excellent live performance, the use of beta-blockers does not impede the attainment of this external goal. If anything, it promotes it. The point here is surely not to test musicians’ ability to overcome stage fright, anxiety, and tremor. On the other hand, the use of beta-blockers would be conceptually problematic in such activities as archery, pistol shooting, snooker, and the like precisely because the point of these activities is (partly) to master control over one’s nerves. The next paragraph discusses similar cases.

In contrast, at least some rules of practice-oriented activities are constitutive. Such rules define which practices are essential to those activities—they stipulate practices within those activities that constitute those activities’ internal goals—and, in abandoning or infringing such rules, one abandons or misses the point of those activities. Constitutive rules define

what practices are central to even engage in those activities, not merely lay out regulations for fair or effective (or whatever else) conduct (Rawls²⁵ Searle²⁶). Constitutive rules impose conceptual limits on the way in which the respective activities may be performed to even count as instances of those activities. Clear examples come from sport. If one shows up at a marathon on roller-skates (Whitehouse et al.²⁹, Schermer²³ and Murray³⁰), one would infringe both regulative and constitutive rules of marathons and so one would do something “wrong” in two different ways. Ethically, one would attempt to get an unfair advantage. Conceptually, one would be missing the point of running a marathon because covering that distance on roller skates is a *different kind of sport*. Crucially, this latter sense of “wrong” (the conceptual wrong) is morally neutral—the roller skater would be doing something wrong in the mere sense of not even doing what the constitutive rules stipulate they must do to even engage in that activity.

Admittedly, one might object that even in mostly practice-oriented activities like sport, many of the constitutive rules are arbitrary. So, why should anyone care about whether the nature of that activity is changed by, for example, allowing roller skates or enhancers to be used? In response, we think that it would be a mistake to suppose that because such rules are arbitrary, nothing would be lost by changing them. For instance, although arbitrary, such rules may nevertheless be imbued with historical or symbolic significance. The marathon has always been exactly 42,195 meters long. And there is a significant historical reason for that: the marathon was instituted in commemoration of the legendary run of the Greek soldier Pheidippides, who, after the Battle of Marathon, covered 42,195 meters to bring the victory message from Marathon to Athens. This is at least a *prima facie* reason to not change such rules. A marathon, as we know it, is about running 42,195 meters.

On the other hand, it is also true that some rules of sport and games are just plainly arbitrary. Apart from respect for a consolidated praxis and possible meanings getting attached to it over time, there are no substantive reasons for having 90-minute football games instead of 86- or 94-minute games. Hence, in the presence of good reasons in favor of change, such rules could perhaps be changed without sacrificing anything important.

Still, many rules in sport and education fall somewhere between these two extremes. They are neither as vital as the ban on wheels in marathons nor as arbitrary as allocating 90 minutes to football games. Some examples might be using wooden versus fiberglass poles in vaulting (Whitehouse et al.²⁹), using manual versus computer-assisted instrumentation in car racing, and using calculators and digital resources at school. We think that an (official) introduction of enhancers into sport and education would be a similarly incompletely arbitrary rule change, and it may therefore be potentially problematic.

COARSE-GRAINED AND FINE-GRAINED DESCRIPTIONS OF ACTIVITIES

Last, how important we take a given constitutive rule to be for a particular activity will also in part depend on just how fine-grained a description of that activity we adopt. As Wittgenstein³¹ argued, no single feature is shared by all games. Arguably, both professional rugby and bouncing a little rubber ball against the wall in front of you while sitting alone at your desk can be called “a game.” Also, the same (in some sense) games can be very different in different times, places, and circumstances. There were car races 60 years ago just as there are today; there are football games between children in many courtyards around the world and there are professional football games; academic research is sometimes conducted using books, pens, and paper and at other times with digital tablets, online research tools, and word processors. On coarse-grained descriptions, 1950’s and today’s car racing are the same game, a children’s courtyard football game is as much a football game as a professional football match, and academic activity of the 1970s is the same as today’s activity. However, on more fine-grained descriptions, these are significantly different activities.⁷ If one considers how much cars have changed—their tires, their embedded technology, how driving techniques have evolved, and how the rules of racing have changed—one will recognize that today’s car racing is not the same sport as 60 years ago. Similarly, once we notice the huge differences in rules, skills, training, stakes, and so forth between professional and children’s football games, how can we maintain that they are identical games? As for academic activity, digital research tools, word processing, and instant electronic communication have drastically reduced the time needed to conduct, write-up, referee, edit, and publish research. Under a fine-grained description, academics today are engaged in a very different kind of activity than their colleagues 30 years ago.

In a nutshell, our point is that cognitive enhancers may be deemed to have a similarly profound effect on a range of intellectual activities if those activities are described in fine-grained detail and if at least some of those details are taken to constitute rather than merely regulate those activities (Cf. Santoni de Sio et al.⁵).

LPI REVISITED

A better rationale for LPI may be offered by looking at sport and education as activities to which constitutive rules and fine-grained descriptions apply.

Consider an athletic activity (*A*) or a learning activity (*L*) in which enhancers (*E*) are not currently used. On a sufficiently fine-grained description, athletes and students who started using *E* would engage in markedly different

activities— A_E and L_E —than the ones for which praise would be distributed to others, namely A and L . They would be engaged in enhanced- A and in enhanced- L . Hence, by taking seriously constitutive rules and a fine-grained description of activities, it is possible to see why enhanced athletes and students may deserve less or even no praise for their accomplishments. It is not that enhancers or enhancer-producers acted in their place. It is not that by enhancing themselves agents became (temporarily) numerically or morally different persons. Finally, it need not even be that the praise that would otherwise be their due needs to be counterbalanced by the unfairness of the competitive advantage that they gained or because they intentionally tried to cheat. Rather, athletes or students who use enhancers would be due less or even no praise because they simply did not engage in the same athletic or learning activity. Because they engaged in *a different activity*, a different yardstick must be used to assess their performance. Praise might still be due to the enhanced athletes and students for A_E and L_E , respectively, but not for A and L simply because they did not engage in those activities.

This is our nature-of-activities justification for LPI, and we think that it allows us to capture the true concerns that lurk beneath the other three justifications while avoiding their drawbacks.

First, there is a grain of truth behind the responsibility-shifting justification; namely, that praise for an activity *under a given description* may indeed be precluded through the use of enhancers. But the claim that praise is diminished *because it is shifted elsewhere* overlooks that praise may still be due to the agent—perhaps even more praise, for all we know—but for the activity *under another description*. For instance, we think that enhanced athletes like Oscar Pistorius still deserve praise.^{vi} In fact, they may display a huge amount of physical and psychological skills in their performance. However, on a fine-grained description, running on prosthetic legs may be a very different sport altogether, and there may even be compelling reasons to resist changing the constitutive and regulative rules of that sport as it currently stands. From this particular standpoint, it may indeed have been a mistake to admit Pistorius into the conventional competition.

The concern for authenticity or personal identity also becomes understandable. By engaging in certain activities, people acquire certain roles. Those who regularly play football in a team *are* football players. Those who enroll in and attend university courses *are* university students. Moreover, by focusing on more fine-grained descriptions of activities, more fine-grained descriptions of the corresponding roles will become relevant. For instance, depending on the kind of team or championship in which they play, football players may be professionals or amateurs. Depending on the course in which they are enrolled, students may be philosophy or medicine students.

Similarly, depending on whether or not they use enhancing substances while preparing for or sitting an exam, language students may either be nonenhanced or enhanced, like Llana from our opening example. We thus propose that the concern about authenticity is wrong-headed to the extent that it suggests that by enhancing themselves, an agent may become a different person or moral subject, one with a separate record of actions and performances. However, it also contains a grain of truth because it points to the fact that the use of a certain enhancing substance can make agents acquire different roles and hence make them open to evaluation qua different kinds of players.

Finally, the nature-of-action justification for LPI has important similarities but also differences with the cheating justification. Both justifications rely on the idea that less praise springs from the agent's violation of a rule of a practice. However, the cheating justification amounts to the idea that enhanced agents deserve no or less praise for their accomplishments because of the counterbalancing effect of an injustice. An implication of this justification is that if LPI depends on cheating, and cheating amounts to unfair advantage over others, then allowing everyone to enhance should defuse LPI. Absent the ban, absent the cheating; and absent the cheating, no ground for LPI. But it seems to us that this retort fails to acknowledge that we may have legitimate grounds to be conservative—to not change the constitutive rules of some activities, but rather to dig our heels in in protest when someone violates those rules—whereas the nature-of-activities justification recognizes this point. Whereas the cheating justification would have us attribute the diminution of praise to the presence of a blemish (intentional cheating or unfair advantage) that must be deducted from the praise that would otherwise be due, in contrast, the nature-of-activity approach claims that less (or even no) praise is due for that activity because the agent engaged in a *different* activity. The *less* praise intuition may actually be a *no* praise intuition. The nature-of-activity justification makes sense of the negative stance against enhanced agents but through the idea of their violation of a constitutive rule of a given practice as that practice currently stands. Furthermore, this justification also explains why changing the rules of the game—for example, simply allowing everyone to use enhancers—won't always suffice to silence moral concerns. Sometimes a rule prohibiting enhancer use may be required to preserve the point of that activity. And those who violate the rule are, in this case, still open to a negative evaluation—even if they are not gaining any unfair advantage—simply because they are *missing the point* of that activity and perhaps because doing so might contribute to water down that activity (at least under a certain fine-grained description).

LPI, Our Nature-of-Activities Justification, and the Enhancement Debate

How does the nature-of-activities justification for LPI bear on the debate about when enhancement should be permitted, encouraged, prohibited, or mandated?

First, on our account, in mainly *practice*-oriented activities—perhaps sports and education—the use of enhancers may present a challenge to the current nature of those activities, and this may potentially expose enhanced agents to legitimate criticism.

Second, our view also entails that people who use enhancers need not necessarily be described as undermining their own agency or personal identity and that they need not be stigmatized as cheaters. Our view leaves open the possibility of a favorable stance on the use of enhancers in sport and education. And although there may indeed be reasons to refrain from altering incumbent constitutive rules of given activities, these reasons need not be so weighty that they can't be overridden by other considerations. For instance, Savulescu et al.²² argue that since we will never eradicate doping in sports, it is better to permit regulated use of enhancers to ensure that athletes' health is adequately monitored. We need not stick to incumbent fine-grained definitions of actual activities at all costs, and, in the long run, the deep-seated features of some activities may gradually erode from pressure due to cultural and social change that eventually makes them into something completely different (Savulescu et al.²²).

We also think that our approach suggests a better methodology for reasoning about this topic than other approaches. First, it marks the distinction between mainly *goal*-directed activities like professions and mainly *practice*-oriented activities like sports (and perhaps education). Moreover, the nature-of-activity approach encourages fine-grained analyses of practice-oriented activities, which makes possible tailored answers to moral questions about enhancement. Not only may we need different rules for the use of enhancing substances by adults and children, but we may also need different rules for different sports (e.g., professional vs. amateur sport) or different kinds of formative activities (e.g., undergraduate vs. graduate exams; exams aimed to assign people to professional positions versus prevalently formative exams like tests internal to a single academic course). We need to think reflectively and reason about what point different activities might have and whether that point would be foregone by allowing enhancers. Each fine-grain defined activity may have its own specific point, this point may have a larger or smaller moral or societal value, and this value may be fostered or jeopardized by enhancers (Santoni de Sio et al.⁵).

An evaluation of the values at stake and the effectiveness of policies aimed at protecting those values must be made before reaching a decision about the permissibility of enhancement in given activities.

Notes

- i. This chapter concerns, therefore, the possible impact of cognitive enhancement on one kind of backward-looking responsibility: praise. For a general discussion on the possible impact of cognitive enhancement on moral and legal responsibility, see Maslen et al.³ For a discussion on the possible impact of cognitive enhancement on forward-looking responsibility (i.e., duties) see also Santoni de Sio et al.⁴
- ii. This methodology has been further developed in Santoni de Sio et al.⁵
- iii. Some may still think that the responsibility-shifting reason for LPI applies at least to some forms of radical enhancement like Oscar Pistorius's amputation of both legs and their replacement with artificial limbs. As a child, Pistorius underwent bilateral leg amputation due to being born without fibulas in each of his legs. He wore artificial prostheses since early childhood, and he mastered them to the point of becoming a successful sprinter. In 2012, he competed in able-bodied Olympics, and although he did not fare as well as he expected, with time, technological advancements will create prosthetic legs that are superior to flesh and bone. At that point, if athletes with prosthetic legs were still allowed to participate in able-bodied races, then, to remain competitive, able-bodied athletes wishing to participate in those races would also need to undergo bilateral leg amputations. Some may think that under such circumstances such athletes should relinquish some praise for their performance because the decisive element would now be a device designed by engineers and implanted by surgeons. However, even in this case, we think that such a responsibility-shifting concern would be based on the unwarranted assumption that artificial limb-equipped athletes would make a less-than-decisive contribution to their achievements in the form of physical and psychological effort and intrinsic talent.
- iv. Also see MacIntyre²⁷ for the distinction between internal and external goods.
- v. On this point, cf. Vincent^{20:318–320} and Vincent³² who discusses the constantly changing nature of activities and the related changes to moral and legal obligations.
- vi. See note viii.

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